

PhD candidate in Artificial Intelligence

Inès BENITO

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EDUCATION

Ecole Polytechnique - HEC Paris

Paris, France

Master of Science Data Science for Business – GPA: 3.94/4

2021 - 2023

- Courses: Probability, Statistics, Linear algebra and regression, Machine Learning, Optimization and Deep learning, Timeseries, NLP, Reinforcement Learning, Causal inference, Interpretability (SHAP, LIME, ICE) and algorithmic fairness, Database management
- Projects: Hydrogen stations deployment plan with the Ministry of Transportation, Time series prediction of bikes, Computer vision for detection of road cracks and use of data augmentation, Detection of medical terms in documents (NER), Banking competition to optimize profits on loans, Business case for green cars leasing using web scraping and NLP analysis, Churn prediction and business case for construction company with the B.C.G.
- Research paper: "A review of Large Language Models: Technical insights to tackle business challenges"

Ecole Nationale Supérieure de Techniques Avancées (ENSTA)

Brest, France

Master of Science in engineering, specialized in Observation Systems and Artificial Intelligence – GPA: 4/4

2018 - 2021

PROFESSIONAL EXPERIENCE

Amazon Web Services (AWS)

Paris, France

Solutions architect for French Public Sector customers

April 2023 – Until now

- Research**
- Supported a research project on timeseries prediction of spiky workloads, utilizing statistical methods, deep learning models, and LLMs. [Paper "Forecasting algorithms for intelligent resource scaling: An experimental analysis" accepted for publication at the ACM Symposium on Cloud Computing \(SoCC 2024\)](#)
 - Developed an automated tool for risk assessment and mitigation in Responsible AI focusing on fairness, transparency, explainability, safety, privacy and security, robustness and veracity, and governance

Teaching

- Developed and delivered a 24-hour 'AI in the Cloud' course for 30 Master's students of the 'Artificial Intelligence, Systems and Data' program at Université Paris Dauphine-PSL, covering AI integration in cloud environments, MLOps, advanced generative AI applications, and led research presentations on LoRA/DoRA for LLM finetuning, LLM architectures (Transformer, Linformer, Mamba, MoE), pruning, quantization and adversarial attacks on LLMs

Customers

- Led the development and optimization of 10+ Generative AI projects for customers, focusing on refining model performance, improving efficiency, and ensuring alignment with business goals. Employed advanced RAG techniques, LLM fine-tuning, and agent-based systems, while implementing robust guardrails to enhance reliability and safety. Leveraged frameworks such as Llama Index and Langchain for efficient data retrieval and processing, and actively contributed to the international AI/ML specialist community by sharing insights and best practices.
- Led end-to-end support for time series prediction project, encompassing model selection and training, architecture design, debugging, and successful deployment into production with MLOps integration

Solutions Architect intern in data science and machine learning

April 2022 (5 months)

- Development of a text-to-speech tool to synthesize emotions by finetuning an open-source pre-trained model using AWS cloud

Thales

Brest, France

Apprentice software engineer in Electronic Warfare, sensitized to protection of secrecy

September 2020 (1 year)

- Simulated electromagnetic environments for electronic warfare detectors in Python

Laboratory of Image Informatics and Information Systems (LIRIS)

Lyon, France

Machine learning researcher intern

May 2020 (4 months)

- Developed and optimized python algorithms for data synthesis using trained LSTM RNNs, which was used to improve the performance of a search engine for handwritten document based on Siamese Neural Network with triplet loss

LANGUAGES, TECHNOLOGY & OTHER

Languages

French (native), English (fluent), Spanish (working knowledge)

Coding/IT

Python, Keras, Tensorflow, Pytorch, Pandas, R, Hadoop, Spark, SQL, Matlab, C
AWS Solutions Architect Professional certification, Linux, Git, Docker